

Multifunctional Wound Dressings That Combine Hemostasis, Antimicrobial Control, and Healing Activity

Executive summary

No single wound dressing design is universally “best” once hemostatic, antimicrobial, and healing-active functions are all required. The strongest conclusion from recent reviews and original papers is that the highest-performing systems are **composite and compartmentalized**, not monolithic: they separate fast-acting hemostatic functions from slower anti-infective and regenerative functions using layered, gradient, core-shell, or trigger-responsive architectures. That design logic appears repeatedly in the recent literature: dual-layer hydrogels, double-layer microneedles, intrinsic antibiofilm hydrogels with low-leaching chemistries, and self-sealing nanofabrics all outperform simpler single-function matrices because wound needs change over time and by wound type. ¹

If the wound is **acute and bleeding**, the most credible design center is still an **adhesive, highly absorbent interface** built from cationic natural biopolymers or protein matrices, optionally reinforced by a proven pro-coagulant such as kaolin or by bioadhesive chitosan chemistry. Kaolin activates Factor XII and accelerates the clotting cascade; chitosan-based systems use charge-based adhesion, platelet activation, coagulation-cascade support, and fibrin-plug formation; collagen and gelatin contribute platelet interaction and fibrin deposition. These are the chemistries that have translated most convincingly into marketed hemostats. ²

If the wound is **infected, chronic, or biofilm-dominated**, the center of gravity shifts. The best systems emphasize **low-leaching intrinsic/contact-active antimicrobials**, ROS/inflammation control, and ECM-like support for re-epithelialization and angiogenesis. A 2024 Nature Communications hydrogel with tethered cationic polyimidazolium plus N-acetylcysteine is notable precisely because it avoids the translation problems of gross antimicrobial leaching and external photothermal hardware while still accelerating closure of MRSA- and carbapenem-resistant *P. aeruginosa* biofilm-infected diabetic wounds in mice. Phase-programmed systems that switch from early antibacterial action to later anti-inflammatory and pro-angiogenic action are especially persuasive for hard-to-heal wounds. ³

Commercial translation remains uneven. Products such as **QuikClot** and **Axiostat** are strong on hemostasis, while **Hydrofera Blue** and **Promogran Prisma** are stronger on infection management and wound-bed optimization. What is mostly *not* available commercially, at least among well-established products with clear evidence and official positioning, is a single frontline product that simultaneously delivers top-tier rapid hemostasis, robust broad-spectrum antibiofilm activity, and stage-matched regenerative signaling. In practice, the market is still dominated by products optimized for one or two of these axes rather than all three. ⁴

The highest-confidence design recommendation, given the user’s unspecified constraints, is therefore an **evidence-based reference architecture**: a **layered or gradient composite dressing** with a rapid

hemostatic/adhesive wound-contact layer, a low-leaching antimicrobial domain, and a regenerative compartment that releases anti-inflammatory and pro-angiogenic cues later. For chronic infected wounds, intrinsic/contact-killing systems and timed anti-inflammatory/angiogenic release look more translationally attractive than antibiotic-heavy dressings or hardware-dependent photothermal systems; for severe traumatic hemorrhage, no regenerative sophistication compensates for inadequate primary hemostasis.

5

Decision framing

Because the wound type, target pathogens, and clinical setting were not specified, this report evaluates dressings against three broad use cases: **acute traumatic bleeding**, **infected acute wounds**, and **infected chronic wounds such as diabetic ulcers**. The literature consistently shows that wound-healing needs vary by phase and by wound class: hemostasis needs dominate early acute care, while chronic wounds require stronger emphasis on biofilm control, inflammation/ROS modulation, angiogenesis, and ECM remodeling. Reviews of hydrogel and modern wound-dressing design also explicitly argue against a one-size-fits-all dressing and in favor of wound-stage-matched, often personalized or adaptive, systems. 6

That framing matters analytically. A dressing optimized for catastrophic hemorrhage can tolerate less moisture-retention sophistication or growth-factor engineering if it stops bleeding reliably within minutes. By contrast, an infected diabetic wound benefits little from a best-in-class trauma hemostat if it cannot disrupt biofilm, control protease/ROS burden, and support keratinocyte migration or angiogenesis. The “best design” therefore depends on which failure mode is most important to prevent first. 7

Material classes and biological mechanisms

Recent reviews converge on a familiar but important materials principle. **Natural polymers** provide biocompatibility, hydration, and tissue-mimetic chemistry; **synthetic polymers** provide control of mechanics, degradation, and processing; **inorganics and nanomaterials** provide rapid hemostasis, antimicrobial potency, or trigger responsiveness; and **composites** are usually necessary to balance all of these properties in one device. Electrospun fiber reviews and hydrogel reviews say the same thing in different ways: natural biopolymers resemble ECM but are often mechanically weak, while synthetic polymers process well but lack native biological cues, so blends and interpenetrating systems dominate serious multifunctional designs. 8

Candidate materials compared

Material class	Representative examples	Hemostatic value	Antimicrobial value	Healing-active value	Main liabilities and best-use note	Representative evidence
Natural polysaccharides	Chitosan, alginate, cellulose, dextran, hyaluronic acid	Chitosan and alginate support rapid hemostasis through charge effects, absorbency, erythrocyte aggregation, and coagulation support; cellulose is highly absorbent.	Chitosan has intrinsic cationic antibacterial activity; hyaluronic acid is usually a carrier rather than a potent antimicrobial by itself.	HA supports hydration, cell migration, angiogenesis, and inflammation modulation; polysaccharides also suit drug loading.	Strong bioactivity but often limited wet strength and variable degradation unless chemically modified or blended.	9
Natural proteins	Collagen, gelatin, silk fibroin	Collagen/gelatin promote platelet interaction and fibrin deposition; silk contributes structural integrity more than direct clot triggering.	Usually weak intrinsic antimicrobial action unless functionalized or loaded.	Strong ECM mimicry, fibroblast adhesion, angiogenesis support, and cell-instructive biology.	Animal-derived immunogenicity, weaker wet adhesion, and sterilization sensitivity can be limiting.	10

Material class	Representative examples	Hemostatic value	Antimicrobial value	Healing-active value	Main liabilities and best-use note	Representative evidence
Synthetic polymers	PEG, PVA, PVP, PU, PCL, PLGA, PNIPAM	Usually indirect; mainly provide shape, swelling control, mechanical support, and drug-release control.	Can host release-based antimicrobials or be converted into contact-killing cationic chemistries.	Excellent for controlled delivery, anti-fouling, and tunable degradation; PEG and PLGA are especially useful as carrier matrices.	Lowest intrinsic bioactivity; often need blending with natural polymers or bioactive fillers.	8
Inorganic hemostats and antimicrobial agents	Kaolin, zeolite, silica, Ag, Cu, ZnO, TiO ₂ , nanozymes, MOFs	Kaolin is clinically validated for Factor XII activation; porous inorganics can absorb fluid and concentrate clotting factors.	Ag, Cu, ZnO, iodine-type systems, nanozymes, and some MOFs provide strong antimicrobial activity through ions, ROS, catalytic effects, or membrane damage.	Some inorganic systems also modulate ROS or oxygen pathways and can support angiogenesis indirectly.	Cytotoxicity, low biodegradability, environmental concerns, and regulatory burden increase with higher release or complex nanoformulations.	11
Carbon and photothermal nanomaterials	Graphene oxide, black phosphorus, CNTs, polydopamine, MXenes	Mostly indirect; can reinforce structure and sometimes improve adhesion.	Enable photothermal or photothermal/photodynamic killing, especially against resistant bacteria and biofilms.	Some can also enable conductivity or oxygen/ROS management relevant to regeneration.	Strongly context-dependent; external light requirement and thermal-safety windows complicate translation.	12

Material class	Representative examples	Hemostatic value	Antimicrobial value	Healing-active value	Main liabilities and best-use note	Representative evidence
Composite systems	Natural/synthetic blends, interpenetrating networks, hybrid hydrogels, core-shell fibers	Best route to combine rapid clotting, wet adhesion, and mechanical integrity.	Best route to combine intrinsic and release-based antimicrobial strategies while controlling leaching.	Best route to stage ECM mimicry, anti-inflammatory action, angiogenesis, and remodeling.	The strongest overall class, but also the most formulation-sensitive and hardest to standardize.	13

Across these materials, the most convincing recurrent pattern is **natural-bioactive plus synthetic-structural plus targeted inorganic/nano-functional additive**. In other words, the best candidates are not “all-natural” or “all-synthetic”; they are *engineered hybrids* that separate jobs among components. ⁸

Hemostatic mechanisms

The hemostatic mechanisms most relevant to multifunctional dressings fall into three broad categories. The first is **physical barrier formation and fluid absorption**: porous or swellable matrices absorb plasma, concentrate clotting factors and blood cells, and form a sealing interface. The second is **pro-coagulant activation**: kaolin is the canonical example, activating Factor XII and accelerating the clotting cascade in commercial QuikClot products. The third is **cellular activation and adhesive clot reinforcement**: chitosan-rich systems use charge-based adhesion with blood components, support platelet activation, promote clotting-cascade activation, and drive fibrin-plug formation; collagen- and gelatin-based systems rely more on platelet interaction and fibrin deposition. ¹⁴

For unspecified wounds, the best hemostatic materials are usually those that combine **fast interfacial sealing** with **mechanical compliance** and **blood compatibility**, because a material that forms clot fragments, leaks at wound edges, or cannot conform to irregular anatomy will underperform even if its chemistry is pro-coagulant. That is precisely why recent self-sealing nanofabrics and bioadhesive hydrogels are drawing attention: they try to solve leakage at the dressing–tissue interface, not only clot speed. ¹⁵

Antimicrobial mechanisms

Four antimicrobial modes are especially relevant to multifunctional wound dressings.

Contact-killing systems use cationic surfaces or tethered antimicrobial groups. Quaternary ammonium-containing polymers act through electrostatic interactions that disrupt membranes and cellular electrolytes, while chitosan’s protonated amino groups destabilize bacterial membranes, interfere with intracellular targets, and chelate essential metal ions. These strategies are attractive because they can reduce the need for high-level antimicrobial release. ¹⁶

Release-based systems deliver antibiotics, metal ions, antiseptics, antimicrobial peptides, or catalytic agents from the matrix. This remains common and powerful, but it raises harder questions about burst

release, depletion, cytotoxicity, and regulatory burden. A common theme in newer work is to reduce leaching while preserving potency. ¹⁷

Photothermal and photodynamic systems use nanomaterials to convert light into heat or ROS. Oxford's 2024 review notes that PTT can be effective against resistant bacteria and biofilms, but also warns that PTT alone may require higher temperatures that risk healthy-tissue damage; hence the field increasingly moves toward multimodal systems rather than PTT monotherapy. The same review documents PTT/PDT combinations achieving high antibacterial effects in representative nanosystems. ¹²

Quorum-sensing disruption is promising but still less mature in wound dressings than contact-killing or release-based designs. A recent ACS review consolidates anti-quorum-sensing materials as an antibiofilm strategy, and wound-specific proof-of-concept work has already embedded quorum-sensing inhibitors in bioactive hydrogels to promote infected wound healing. Analytically, this looks most attractive as an **antibiotic-sparing antibiofilm adjunct**, especially where AMR concerns make classic antibiotic-loaded dressings less favorable. ¹⁸

Healing-active mechanisms

The literature uses “healing-active” to mean more than passive moisture balance. Three high-value mechanisms recur.

The first is **biological cargo delivery**, including growth factors, proteins, vesicles, or pro-angiogenic small molecules. Hydrogels are especially favored here because they can serve as targeted delivery vehicles for drugs, proteins, or cells and because their structure resembles ECM. Double-layer microneedles and controlled-release hydrogel systems show the same principle in different forms: rapid early anti-infective release, followed by later pro-angiogenic or anti-inflammatory signaling. ¹⁹

The second is **ECM mimicry and cell recruitment**. Natural polymers such as collagen, gelatin, silk fibroin, soy-protein-containing fibers, and HA-rich hydrogels support fibroblast adhesion, keratinocyte migration, angiogenesis, and ECM remodeling. In practical design terms, high-surface-area nanofibers and hydrated 3D hydrogels are the leading ECM-mimetic formats. ⁸

The third is **microenvironment reprogramming**, especially anti-inflammatory and ROS-modulating action. This is critical in chronic diabetic and infected wounds, where excess ROS, inflammatory cytokines, and protease burden trap the wound in a non-healing state. Programmed hydrogels that switch from an early antibacterial phase to a later ROS-scavenging, angiogenesis-supporting phase directly target that problem. Antimicrobial microneedle patches carrying adipose-tissue-derived apoptotic vesicles extend the concept further by supporting antibacterial control while also promoting rapid appendage regeneration, with mature regularly arranged hair follicles reported as early as day 8 and visible hair by day 16 in infected wounds. ²⁰

Architectures and integration strategies

Form factor is not a packaging decision; it is a mechanistic decision. Sponges, hydrogels, films, nanofibers, microneedles, and multilayer patches each bias the dressing toward different combinations of hemostasis,

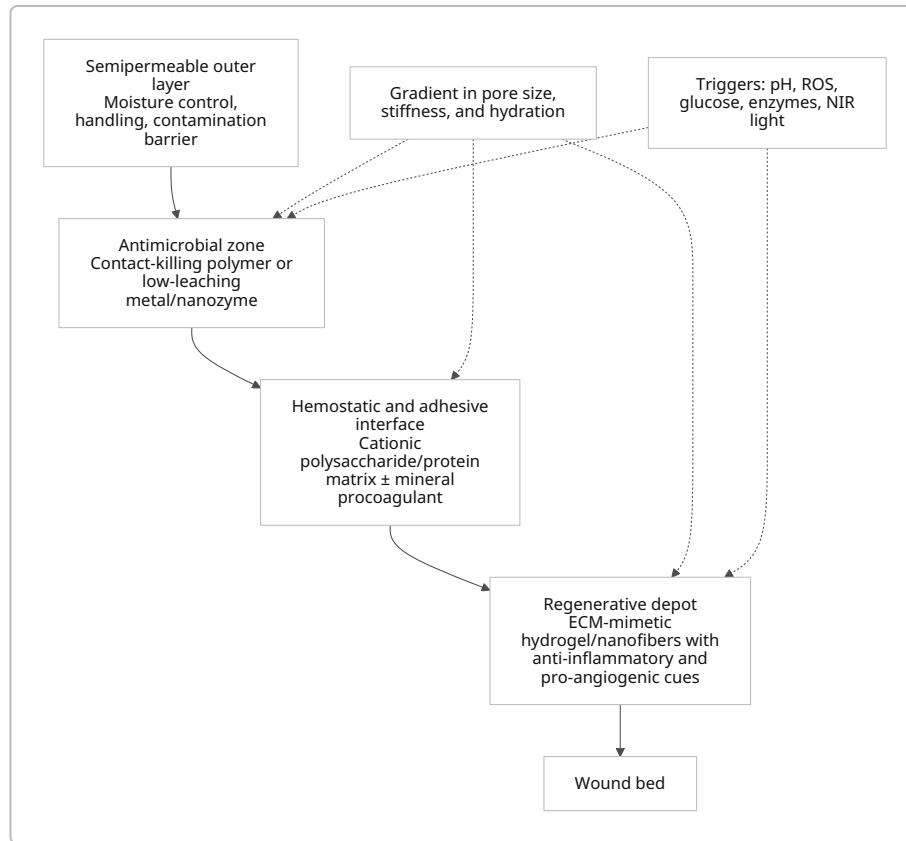
infection control, conformability, exudate handling, and regenerative signaling. Reviews of hydrogels and fibrous dressings consistently show that architecture is often as important as chemistry. ²¹

Form factors mapped to function

Form factor	Hemostasis potential	Antimicrobial potential	Healing-active potential	Best-use pattern and limitations	Evidence
Sponges and foams	High for compressible, absorbent blood uptake and rapid interfacial sealing.	Moderate to high when loaded or intrinsically antimicrobial.	Moderate; mainly support moisture/exudate handling and wound-bed preparation.	Very practical for exudative or bleeding wounds, but less precise for staged cargo delivery than multilayer or microneedle systems.	²²
Hydrogels	Moderate to high when adhesive and fast-gelling; especially good on irregular surfaces.	High through intrinsic, release-based, or trigger-responsive antimicrobial functions.	High because of ECM-like hydration, cell/drug delivery, anti-inflammatory and angiogenic programming.	Best all-around platform for chronic and infected wounds, but pure hydrogels may need reinforcement for heavy bleeding or mechanical abuse.	²³
Films	Low to moderate as stand-alone hemostats.	Moderate, especially for surface antisepsis or barrier function.	Moderate when used as protective semipermeable outer layers.	Strong for shallow wounds and moisture control, weaker for deep exudative or irregular wounds.	²⁴
Electrospun nanofibers	Moderate; strong surface area and clot interaction, especially when made adhesive or bioactive.	High, including antibiotics, metals, AMPs, and contact-active chemistries.	High due to ECM-mimicry, porosity, gas exchange, and controlled release.	Excellent for chronic wounds and regenerative designs; scale-up is improving but manufacturing consistency still matters.	²⁵

Form factor	Hemostasis potential	Antimicrobial potential	Healing-active potential	Best-use pattern and limitations	Evidence
Microneedles	Low as primary external hemostats, but high for targeted local delivery into wound tissue or biofilms.	High for deep delivery of antibiotics, AMPs, vesicles, or prodrugs.	High for tissue-targeted angiogenic and regenerative signaling.	Most useful for infected diabetic or biofilm-laden wounds rather than exsanguinating trauma.	26
Patches	Variable; depends on backing, adhesive, and active layer design.	Variable to high.	Variable to high.	Good platform for combining handleability and layered functionality.	27
Layered or gradient structures	Very high system-level potential because each zone can be optimized separately.	Very high, especially when early and late actions are decoupled.	Very high, especially for staged anti-inflammatory and pro-angiogenic release.	Best architecture for multifunctionality, but fabrication and quality control become more demanding.	28

The most rigorous recent multifunctional systems use one of three integration strategies. The first is **spatial compartmentalization**: different active ingredients are kept in separate layers or domains so they do not quench one another or release in the wrong phase. The second is **temporal programming**: an early-phase antibacterial or hemostatic component gives way to a later anti-inflammatory or angiogenic component. The third is **stimuli-responsive release**: pH, ROS, glucose, temperature, enzymes, or NIR light trigger a change in function or payload release. Dual-layer hydrogels, double-layer microneedles, and light-triggered AMP microneedles are strong concrete examples of all three approaches. 29



From a design standpoint, this layered or gradient logic is the most defensible answer to the multifunctionality problem. It avoids forcing one chemistry to do everything at once, and it aligns better with the wound-healing timeline documented across modern hydrogel and microneedle studies. ³⁰

Performance evaluation and translational constraints

A multifunctional dressing should not be judged by a single headline result. The minimum serious test package spans **hemostasis, antimicrobial efficacy, biocompatibility, healing efficacy, mechanics and mass transport**, and **degradation/sterility stability**. Hemostatic studies commonly report time to hemostasis, total blood loss, clotting index, platelet/blood-cell adhesion, and survival in liver, kidney, femoral, or carotid bleeding models. Antimicrobial studies should move beyond planktonic kill percentages toward pathogen-specific log reduction, biofilm eradication or disruption, and durability of action. Healing studies should report closure kinetics together with histology, re-epithelialization, angiogenesis, collagen deposition, inflammatory markers, and sometimes appendage regeneration. ³¹

For physical performance, EN 13726 remains central in Europe for absorbency, moisture-vapor transmission, waterproofness, and extensibility, and NHS/SMTL resources explicitly identify EN 13726 as the core standardized framework for these dressing characteristics. Microbial barrier testing also has a dedicated ASTM method, ASTM E3383-24, and EN 17854:2024 has been introduced for antimicrobial wound-dressing claims. In practice, that means developers should benchmark not only whether a dressing kills bacteria, but whether it still provides fluid handling, breathability, and barrier integrity at clinically relevant loading. ³²

The translational lesson is straightforward: a dressing that performs brilliantly in one assay but degrades mechanically, leaks, macerates tissue, or loses function after sterilization is not a winning design. The 2024 mupirocin-loaded Soluplus/soy-protein nanofiber paper is a good example of useful breadth in reporting: water absorption, WVTR, tensile strength, degradation, release kinetics, bacterial killing, cell compatibility, and in vivo healing were all reported together rather than selectively. ³³

Sterilization is a major but underappreciated filter in this field. Reviews and original studies show that **steam can collapse alginate-based structures, ethylene oxide can change swelling and water content in PVA-based constructs, radiation can both sterilize and in some systems crosslink hydrogels**, and nanofibrous materials can change morphology or performance depending on the sterilization route. In other words, sterilization is not an afterthought; it is part of formulation design. A dressing that depends on delicate porous structure, hydrogen bonding, or loaded biologics must be designed around a validated sterilization strategy from the outset. ³⁴

Scalability introduces another set of trade-offs. Electrospinning is attractive because it can generate ECM-like fibers with high surface area and can now be scaled by multi-needle or nozzleless approaches, but large-scale production still has to manage solvent handling, uniformity, throughput, and quality control. Reviews also note that electrospinning is compelling partly because of its relative ease of production and scale-up compared with some more exotic fabrication routes. By contrast, simple hydrogels, foams, and films are often easier to manufacture reproducibly, while systems containing living components, vesicles, or fragile biologics face added stability and CMC burdens. ³⁵

Regulation is becoming more important, not less. FDA states that it **proposed in 2023** to classify certain wound dressings and wound washes containing medically important antimicrobials into class III because of antimicrobial-resistance concerns, and FDA separately emphasized these AMR concerns again in 2026. In addition, biological evaluation for wound dressings sits within the ISO 10993-1 risk-management framework, which links biocompatibility testing to the device's materials, design, and nature/duration of body contact. In Europe, MDCG 2021-24 rev.1 was updated in April 2026 and continues to guide classification of wound dressings under the MDR. The practical implication is clear: **antibiotic-sparing, low-leaching, device-centric antimicrobial strategies may be easier to advance than heavily antibiotic-dependent dressings**, especially when long-term AMR risk is hard to justify. ³⁶

Representative prototypes and commercial products

The strongest recent prototypes are notable not because they contain the most ingredients, but because they use architecture to stage functions intelligently.

Recent prototypes

Prototype	Form factor and design logic	Functional package	Key reported data	Why it matters
Gao et al., 2023, DMN@TCH/DFO	Double-layer microneedle with HA tip carrying tetracycline hydrochloride and chitosan/silk-fibroin substrate carrying deferoxamine.	Early antibacterial release plus sustained pro-angiogenic support.	Adequate mechanical properties, excellent swelling and biocompatibility; reduced inflammatory responses, promoted angiogenesis and collagen deposition, and accelerated diabetic wound healing in rats. ³⁷	Strong example of spatial/temporal separation of anti-infective and healing-active tasks.
Su et al., 2023, NIR-triggered AMP microneedles	TD-coated PVP microneedles loaded with IR780 and antimicrobial peptide W379.	Triggered peptide release; antibiofilm therapy.	Near-infrared light melted the phase-change coating to release peptide; the system showed high antibacterial efficacy in vitro and excellent antibiofilm activity ex vivo and in vivo. ³⁸	Excellent proof of concept for on-demand therapy, especially for biofilm-heavy wounds.
Pranantyo et al., 2024, PPN hydrogel	Synthetic hydrogel with tethered cationic polyimidazolium and N-acetylcysteine.	Intrinsic antibiofilm plus antioxidant activity with low leaching.	Accelerated closure of MRSA- and carbapenem-resistant <i>P. aeruginosa</i> biofilm-infected diabetic wounds in mice; adaptable to flat and deep wounds without photothermal hardware. ³⁹	One of the clearest translational arguments for intrinsic, low-leaching antibiofilm chemistry.

Prototype	Form factor and design logic	Functional package	Key reported data	Why it matters
Peng et al., 2024, PTMH	Dual-layer hydrogel with SA/ZnO lower layer and PNIPAM/Cu5.4O upper layer.	Early antibacterial ROS generation, later ROS scavenging, anti-inflammatory action, and angiogenesis.	The lower layer attacked bacteria under visible light, then degraded; later Cu5.4O release scavenged excess ROS, neutralized inflammatory factors, and enhanced angiogenesis in infected diabetic wounds. ⁴⁰	Strong demonstration of time-programmed wound-microenvironment regulation.
Jahani et al., 2024, Sol/SPI/Mp nanofibers	Electrospun Soluplus/soy-protein-isolate nanofibers loaded with mupirocin.	Antibacterial plus ECM-like support and drug delivery.	Water absorption $300.83 \pm 29.72\%$; WVTR 821.8 ± 49.12 g/m ² ·day; degradation $33.73 \pm 3.55\%$ after 5 days; tensile strength 3.61 ± 0.29 MPa; $85.90 \pm 6.02\%$ drug release in 5 days; killed $93.06 \pm 5.40\%$ of <i>S. aureus</i> and $90.40 \pm 5.66\%$ of <i>E. coli</i> ; biocompatible with HaCaT cells; fastest healing in rat wounds. ³³	Rare paper with a broad and useful performance package, not only a healing image series.
Ma et al., 2024, ApoEVs-AT@MNP	HAMA-based soluble microneedle tips carrying adipose-tissue apoptotic vesicles on an antibacterial PLMA/HAMA substrate.	Antibacterial action plus high-quality regenerative signaling.	Infected-wound healing was promoted while bacterial proliferation was inhibited; mature, regularly arranged hair follicles were seen by day 8, and visible hair by day 16. ⁴¹	A strong example of healing-active functionality going beyond closure to functional regeneration.

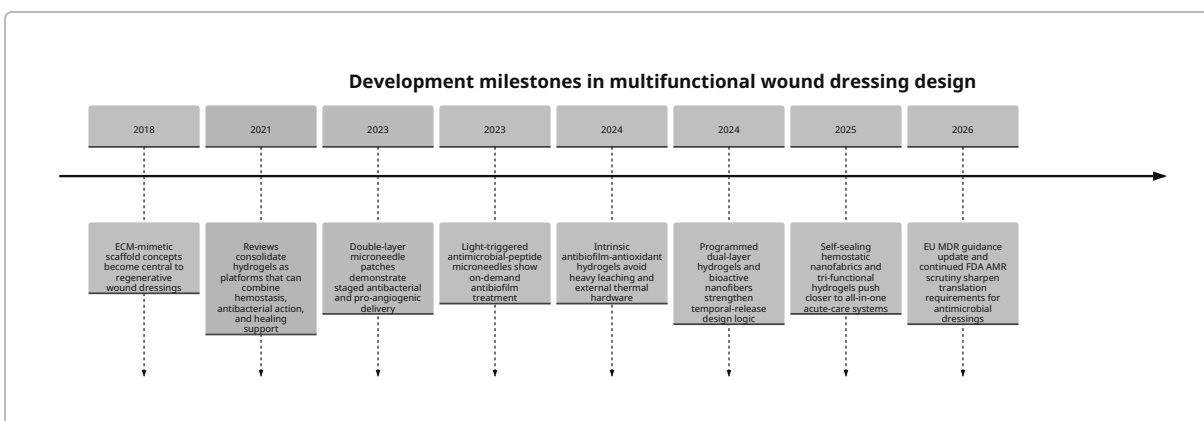
Prototype	Form factor and design logic	Functional package	Key reported data	Why it matters
Zhou et al., 2025, HA@TA-Okra	HAMA hydrogel integrated with tannic acid and okra extract.	Antibacterial, antioxidant, and hemostatic.	Rapid gel response, swelling regulation, antioxidant ability; significantly outperformed conventional dressings in rat-hemostasis testing and improved infected wound repair with reduced inflammation. ⁴²	Shows the growing appeal of plant-derived multifunctional chemistry for simplified tri-function designs.
Fang et al., 2025, self-sealing nanofabric	Anionic/cationic electrospun nanofabric that self-seals on blood contact and adheres strongly to tissue.	Physical hemostatic barrier plus thrombus reinforcement, antibacterial activity, wound healing promotion.	In pig femoral artery injury, blood loss was reported as only about 8% of the loss seen with Combat Gauze; the nanofabric also showed good toughness, flexibility, biodegradability, hemocompatibility, and cytocompatibility. ⁴³	Among the best current examples of solving edge leakage and directional blood seepage.

Commercial products

Product	Official design features	Primary strengths	Representative performance or claim	Analytical takeaway
QuikClot Combat Gauze / Control+	Kaolin-impregnated nonwoven; kaolin activates Factor XII and accelerates the clotting cascade. ⁴⁴	Fast hemostasis in trauma and procedural bleeding.	Official site cites 95% successful hemostasis in a Mayo Clinic prehospital study and 91% in an Israeli Defense Forces series; a 2023 study reported 79.1% hemostasis within 5 min for QuikClot Control+ versus 58.4% control. ⁴⁵	Benchmark hemostat, but not a healing-active platform.

Product	Official design features	Primary strengths	Representative performance or claim	Analytical takeaway
Axiostat	100% chitosan hemostatic gauze/patch; official mechanism includes charge-based blood adhesion, platelet activation via TLR, clotting-cascade activation, and fibrin-plug formation; official product pages also claim bacterial barrier function. ⁴⁶	Rapid hemostasis with bioadhesive chitosan chemistry.	Official publications page reports >80% lower post-treatment blood loss than kaolin gauze in a swine study and 100% survival; clinical summaries report mean hemostasis times around 4.68–5.75 min in different settings. ⁴⁷	Strong hemostatic design with some antimicrobial positioning, but limited explicit healing-active function.
Hydrofera Blue	Antibacterial PVA/PU foam with organic pigments; capillary-wicking “natural negative pressure,” broad-spectrum antibacterial protection, biofilm reduction support, and moist-healing environment. ⁴⁸	Bioburden control, exudate management, wound-bed optimization.	Official site states the foam can hold 12 times its weight in exudate, supports broad-spectrum antibacterial protection, and reported 90% epibole flattening at two weeks versus 25% for silver dressings in a cited comparison. ⁴⁹	Strong antimicrobial/healing-support product, but not a primary trauma hemostat.
Promogran Prisma	Collagen + oxidized regenerated cellulose + silver matrix; positioned to address inflammation and bioburden and modulate protease activity. ⁵⁰	Healing-active wound-bed modulation with antimicrobial support.	Official page states 1.74× higher odds of complete healing and cites RCTs/meta-analyses; ORC+collagen has a stronger in vitro effect than collagen alone on reducing MMP and elastase activity. ⁵⁰	Strong healing-active and antimicrobial-support dressing, but not an acute hemostat.

The commercial comparison makes the current gap obvious. The market has credible **hemostatic specialists** and credible **infection/healing specialists**, but relatively few mature products that fuse all three pillars at comparable strength. That gap helps explain why current research prototypes emphasize layered or stage-programmed integration rather than simple incremental improvement of existing single-purpose formats. ⁴



This milestone sequence captures the field's direction: from single-material dressings, to multifunctional hydrogels and fibers, to **architecturally programmed systems** that aim to match the chronology of wound repair. Regulatory pressure on antimicrobial claims is now reinforcing that trend by favoring designs that can justify their anti-infective function without creating avoidable AMR burdens. ⁵¹

Design recommendations and research gaps

For unspecified wounds, the best evidence supports an **evidence-based reference product profile** rather than a single exact formula. The most robust profile is a **multicomponent, layered or gradient device** with four coordinated traits: a wound-contact zone that achieves immediate hemostasis and wet adhesion; a low-leaching antimicrobial domain that works against planktonic bacteria and biofilm; an ECM-mimetic regenerative matrix that supports re-epithelialization and angiogenesis; and a temporal program that reduces inflammation/ROS after the infection-control phase. This recommendation is partly a synthesis rather than a direct claim from any one paper, but it is strongly supported by the repeated success of dual-layer hydrogels, double-layer microneedles, intrinsic antibiofilm hydrogels, and self-sealing hemostatic nanofabrics. ⁵²

Material selection should follow workload. For **hemostatic-first** applications, chitosan-, collagen-, or gelatin-containing composites with proven adhesive or pro-coagulant mechanisms remain the most defensible. For **infection-first chronic wounds**, low-leaching contact-active systems, antibiofilm strategies, and anti-inflammatory/antioxidant components deserve priority over simply loading more antibiotic. For **regeneration-first** designs, ECM mimicry and temporally staged angiogenic signaling are more persuasive than single-shot growth factor loading, which often introduces stability and sterilization difficulty. ⁵³

Two translational recommendations stand out. First, developers should **design backward from sterilization and regulation**, not forward from bench efficacy. If a formulation cannot tolerate gamma, e-beam, or EO, or if its antimicrobial claim inadvertently pulls it into a more difficult regulatory pathway, the smartest chemistry on paper may still be the wrong product strategy. Second, developers should favor **device-led antimicrobial mechanisms** where possible: intrinsic cationic surfaces, tethered antimicrobials, catalytic ROS control, biofilm-disruptive architectures, or quorum-sensing interference may prove more durable strategically than broad antibiotic release. ⁵⁴

The main research gaps are now less about discovering another active ingredient and more about **standardized, comparative, clinically relevant validation**. The field still needs more head-to-head studies

against commercial comparators, more polymicrobial and biofilm-rich wound models, clearer long-term biosafety data for nanomaterials and photothermal systems, sterilization-compatible stabilization methods for growth factors or vesicles, and scalable manufacturing methods for multilayer and microneedle devices. Reviews of hydrogel wound dressings also explicitly note the lack of fully wound-stage-comprehensive systems and the ongoing need for personalized, monitorable dressings that can adapt to real wound trajectories rather than idealized models. ⁵⁵

The most important gap, however, is conceptual. Many papers still claim success after showing some combination of bacterial reduction, collagen deposition, or faster closure, but relatively few studies **prove that their integrated functions are truly synergistic rather than merely co-located**. The next generation of best-in-class studies should therefore test whether hemostatic chemistry impairs later regeneration, whether antimicrobial modules raise long-term cytotoxicity or regulatory complexity, and whether timed or compartmentalized designs genuinely outperform simpler systems under the same test conditions. Until that standard becomes commonplace, the best designs are not the most feature-rich ones, but the ones whose multifunctionality is **mechanistically separated, quantitatively benchmarked, sterilization-compatible, and realistically translatable**. ⁵⁶

¹ ¹⁹ ²³ ³⁰ ⁵¹ ⁵⁵ ⁵⁶ <https://pubs.rsc.org/en/content/articlehtml/2024/ma/d3ma00682d>
<https://pubs.rsc.org/en/content/articlehtml/2024/ma/d3ma00682d>

² ⁴ ⁷ ¹¹ ¹⁴ ⁴⁴ ⁴⁵ <https://quikclot.com/>
<https://quikclot.com/>

³ ³⁹ <https://www.nature.com/articles/s41467-024-44968-y>
<https://www.nature.com/articles/s41467-024-44968-y>

⁵ ²⁰ ²⁸ ²⁹ ⁴⁰ ⁵² <https://link.springer.com/article/10.1186/s12951-024-02390-y>
<https://link.springer.com/article/10.1186/s12951-024-02390-y>

⁶ ²¹ ²⁷ <https://www.mdpi.com/2310-2861/11/2/123>
<https://www.mdpi.com/2310-2861/11/2/123>

⁸ ²⁴ <https://www.mdpi.com/2076-3417/15/4/1725>
<https://www.mdpi.com/2076-3417/15/4/1725>

⁹ ¹⁰ ⁵³ <https://www.the-innovation.org/article/doi/10.59717/j.xinn-mater.2025.100164>
<https://www.the-innovation.org/article/doi/10.59717/j.xinn-mater.2025.100164>

¹² ¹⁶ ¹⁷ <https://academic.oup.com/burnstrauma/article/doi/10.1093/burnst/tkae046/7916379>
<https://academic.oup.com/burnstrauma/article/doi/10.1093/burnst/tkae046/7916379>

¹³ ²⁵ <https://www.mdpi.com/2079-6439/13/8/100>
<https://www.mdpi.com/2079-6439/13/8/100>

¹⁵ ⁴³ <https://www.nature.com/articles/s41467-025-60244-z>
<https://www.nature.com/articles/s41467-025-60244-z>

¹⁸ <https://pubs.acs.org/doi/10.1021/acsami.4c01428>
<https://pubs.acs.org/doi/10.1021/acsami.4c01428>

²² <https://atm.amegroups.org/article/view/66749/html>
<https://atm.amegroups.org/article/view/66749/html>

26 37 <https://pubs.rsc.org/en/content/articlelanding/2023/bm/d2bm01588a>

<https://pubs.rsc.org/en/content/articlelanding/2023/bm/d2bm01588a>

31 <https://pmc.ncbi.nlm.nih.gov/articles/PMC8445204/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8445204/>

32 <https://standards.iteh.ai/catalog/standards/cen/041555b3-d2e5-4702-9a91-3589c000c27e/en-13726-2023?srsId=AfmBOopl9oezObcW4MjAWuSDAPhfGj8gK23DnzYWQAw-jNgz97aHBj4>

[https://standards.iteh.ai/catalog/standards/cen/041555b3-d2e5-4702-9a91-3589c000c27e/en-13726-2023?](https://standards.iteh.ai/catalog/standards/cen/041555b3-d2e5-4702-9a91-3589c000c27e/en-13726-2023?srsId=AfmBOopl9oezObcW4MjAWuSDAPhfGj8gK23DnzYWQAw-jNgz97aHBj4)

[srsId=AfmBOopl9oezObcW4MjAWuSDAPhfGj8gK23DnzYWQAw-jNgz97aHBj4](https://standards.iteh.ai/catalog/standards/cen/041555b3-d2e5-4702-9a91-3589c000c27e/en-13726-2023?srsId=AfmBOopl9oezObcW4MjAWuSDAPhfGj8gK23DnzYWQAw-jNgz97aHBj4)

33 Antibacterial and wound healing stimulant nanofibrous dressing consisting of soluplus and soy protein isolate loaded with mupirocin | Scientific Reports

<https://www.nature.com/articles/s41598-024-78161-4>

34 Alginate in Wound Dressings - PMC - NIH

https://pmc.ncbi.nlm.nih.gov/articles/PMC6027439/?utm_source=chatgpt.com

35 <https://www.mdpi.com/1999-4923/13/2/286>

<https://www.mdpi.com/1999-4923/13/2/286>

36 <https://www.fda.gov/about-fda/economic-impact-analyses-fda-regulations/medical-devices-general-and-plastic-surgery-devices-classification-certain-solid-wound-dressings>

<https://www.fda.gov/about-fda/economic-impact-analyses-fda-regulations/medical-devices-general-and-plastic-surgery-devices-classification-certain-solid-wound-dressings>

38 [https://www.researchgate.net/publication/](https://www.researchgate.net/publication/368926073_Triggered_release_of_antimicrobial_peptide_from_microneedle_patches_for_treatment_of_wound_biofilms)

[368926073_Triggered_release_of_antimicrobial_peptide_from_microneedle_patches_for_treatment_of_wound_biofilms](https://www.researchgate.net/publication/368926073_Triggered_release_of_antimicrobial_peptide_from_microneedle_patches_for_treatment_of_wound_biofilms)

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/368926073_Triggered_release_of_antimicrobial_peptide_from_microneedle_patches_for_treatment_of_wound_biofilms)

[368926073_Triggered_release_of_antimicrobial_peptide_from_microneedle_patches_for_treatment_of_wound_biofilms](https://www.researchgate.net/publication/368926073_Triggered_release_of_antimicrobial_peptide_from_microneedle_patches_for_treatment_of_wound_biofilms)

41 <https://link.springer.com/article/10.1186/s12951-024-02845-2>

<https://link.springer.com/article/10.1186/s12951-024-02845-2>

42 <https://link.springer.com/article/10.1186/s12951-025-03148-w>

<https://link.springer.com/article/10.1186/s12951-025-03148-w>

46 <https://axiostat.com/technology/>

<https://axiostat.com/technology/>

47 <https://axiostat.com/publications/>

<https://axiostat.com/publications/>

48 49 <https://hydrofera.com/why-blue/>

<https://hydrofera.com/why-blue/>

50 <https://www.solventum.com/en-gb/home/medical/advanced-wound-care/advanced-wound-dressings/promogran/>

<https://www.solventum.com/en-gb/home/medical/advanced-wound-care/advanced-wound-dressings/promogran/>

54 Effect of Ethylene Oxide Sterilization on Polyvinyl Alcohol ...

https://pmc.ncbi.nlm.nih.gov/articles/PMC7580577/?utm_source=chatgpt.com